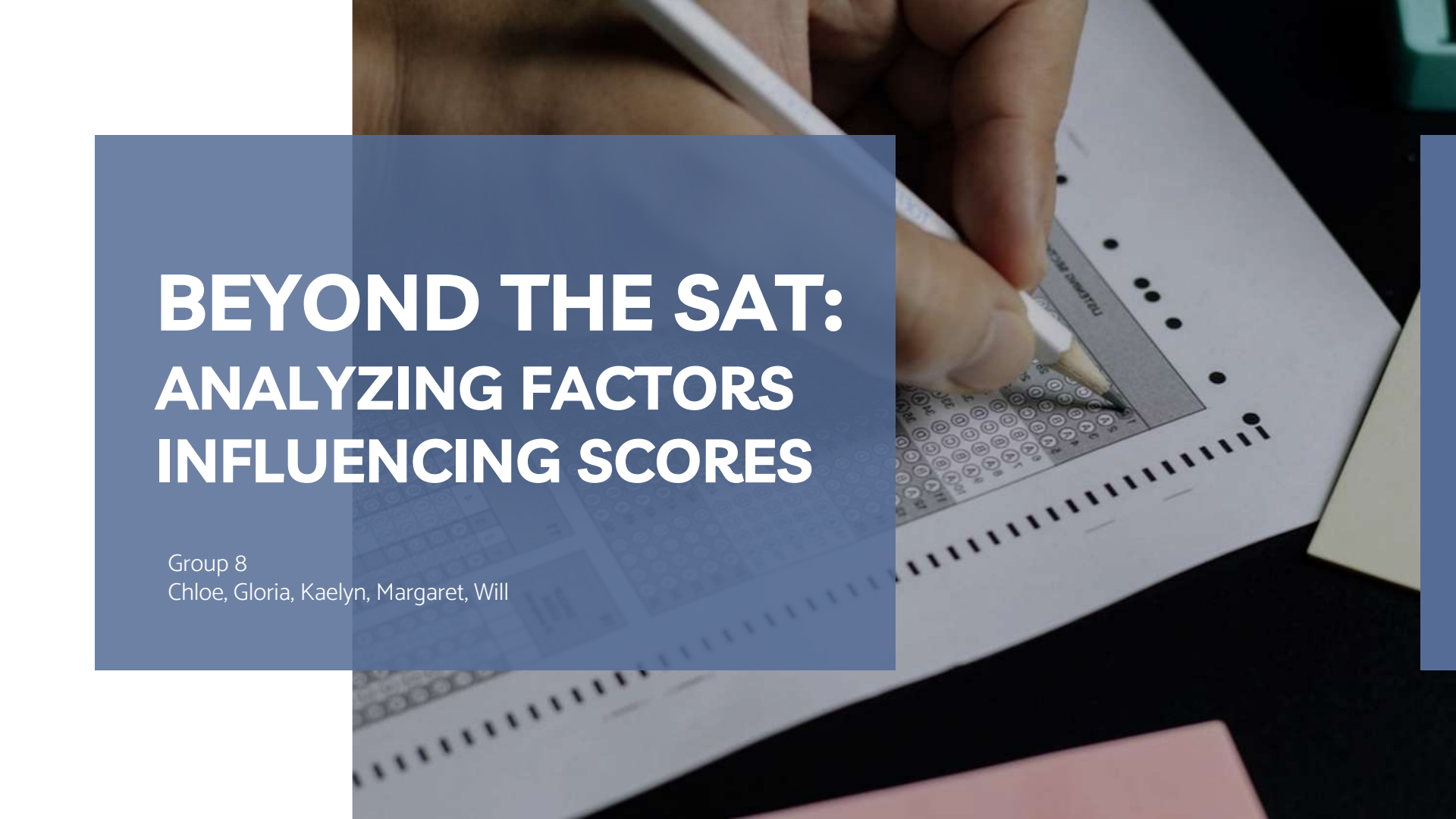


The background of the slide is a close-up photograph of a person's hand using a white pencil to fill out a SAT answer sheet. The hand is positioned in the upper right, with the pencil tip touching one of the bubbles on the grid. The answer sheet is white with a grey grid of bubbles. To the right of the grid, there are several black dots and a series of vertical lines. The entire image is partially covered by a semi-transparent blue rectangle on the left side, which contains the title and group information.

BEYOND THE SAT: ANALYZING FACTORS INFLUENCING SCORES

Group 8

Chloe, Gloria, Kaelyn, Margaret, Will

Table of Contents

**Context &
Hypothesis**

1

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**About the
Dataset**

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**Findings &
Conclusion**

1. CONTEXT & HYPOTHESIS

Why is the SAT important?

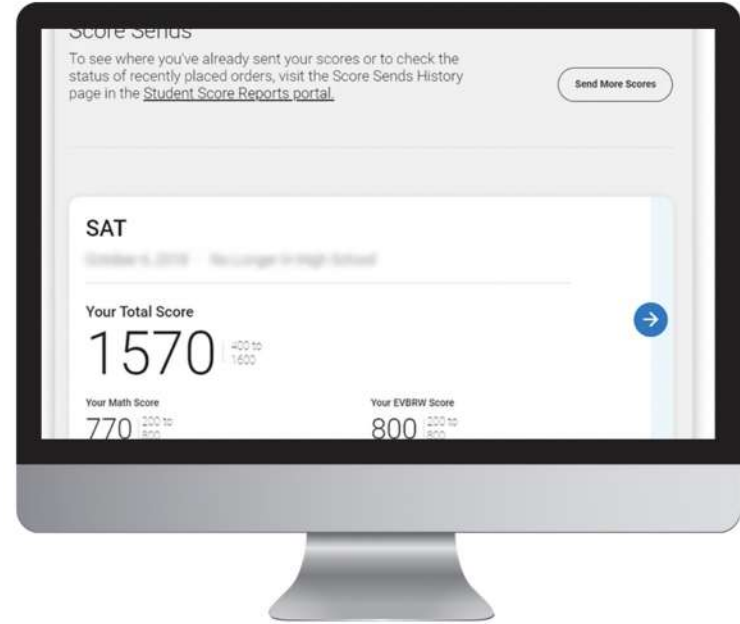
**Heavily
weighted in
college
admissions**

**Measure of
pure academic
capacity**

**Considered a
standardized
exam fair for
all students**

Question:

Besides, pure
academic capacity,
are there **other**
variables affecting
SAT scores?



YES!

Yes. Here's why.

Hypothesis

1. We predict that SAT is not a baseline **measure of pure academic merit**

2. Rather, factors such as district expenditure, socioeconomic background, and cultural factors allow students to **access additional test preparation or resources** which allows them to obtain higher scores.

Factors which may affect SAT scores:



Cultural Bias



**Socioeconomic
background**



**Access to Quality
Education (i.e. school
expenditure)**

2. ABOUT THE DATASET

Why we chose our dataset?

(“MA_Public_Schools_2017.csv”)



Percentage of gender,
ethnicity, school spending,
economic situation, SAT score
of public schools in
Massachusetts.

Credibility

- Found on Kaggle
- Recent dataset (2017)

Relevance

Includes relevant variables (gender,
wealth, school spending) to verify
hypothesis

Original Dataset: Limitations



(1) Only Massachusetts



**(2) Old SAT
(Out of 2400)**



(3) Only Public Schools

3. DATASET MANIPULATION

(Cleaning, EDA, Model)

Cleaning the Dataset

**1. Remove elementary and middle schools
(no SAT score)**

2. Remove irrelevant variables (i.e. AP scores)

3. Rename variables for simplicity

4. Converted old SAT score to new SAT score to remove limitations

EDA: Exploratory Data Analysis

1

Histogram

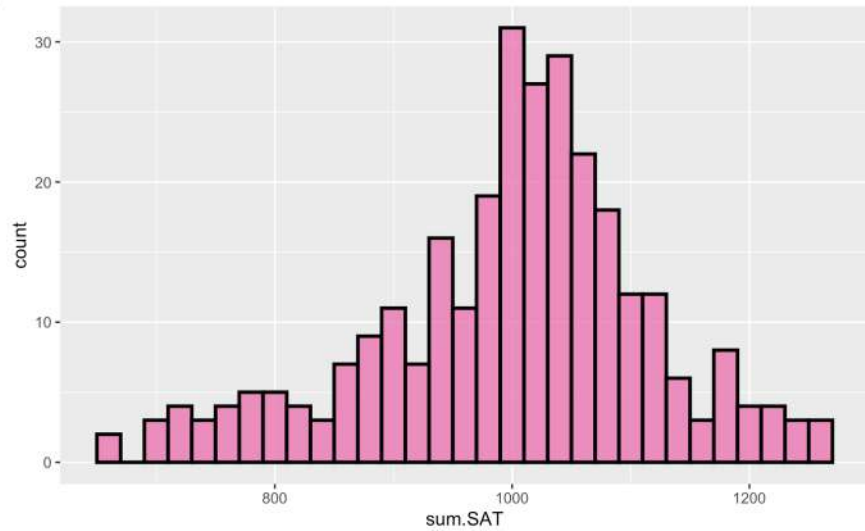
2

Boxplot

3

Heatmap

Exploratory Data Analysis

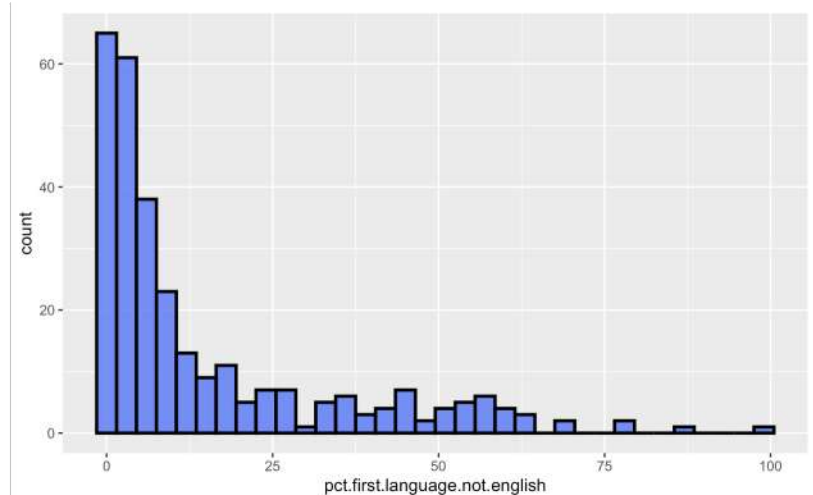


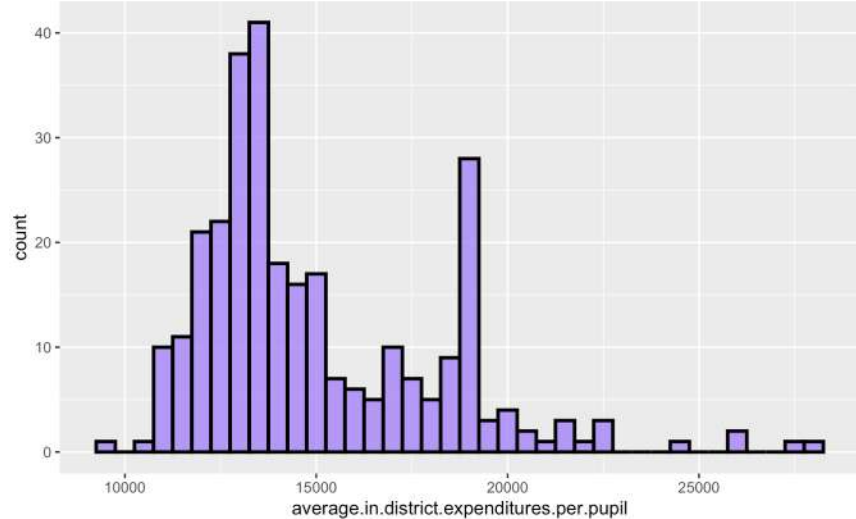
1. Distribution of SAT Scores

- Normal distribution (bell-shaped curve)

2. % of People Whose First Language Isn't English

- Heavily skewed to the right
- Few schools have >75%



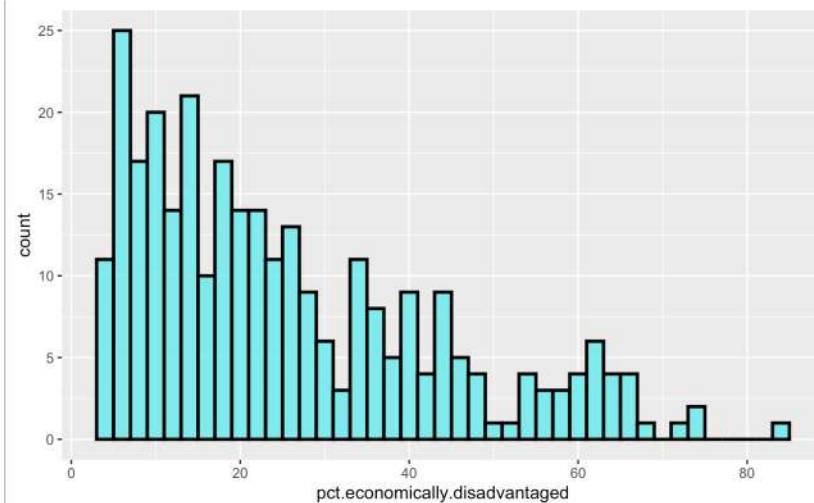


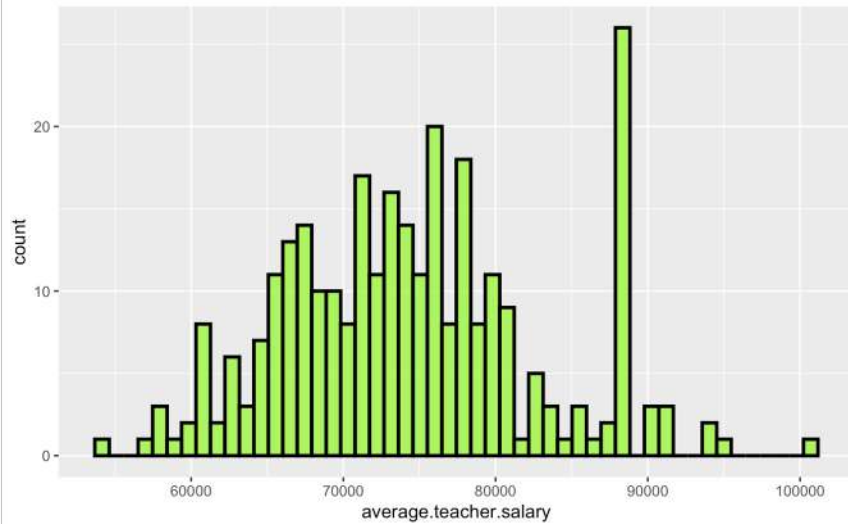
3. Average In District Expenditures Per Pupil

- Noticeable Outlier

4. % of People Who are Economically Disadvantaged

- Clear negative linear trend



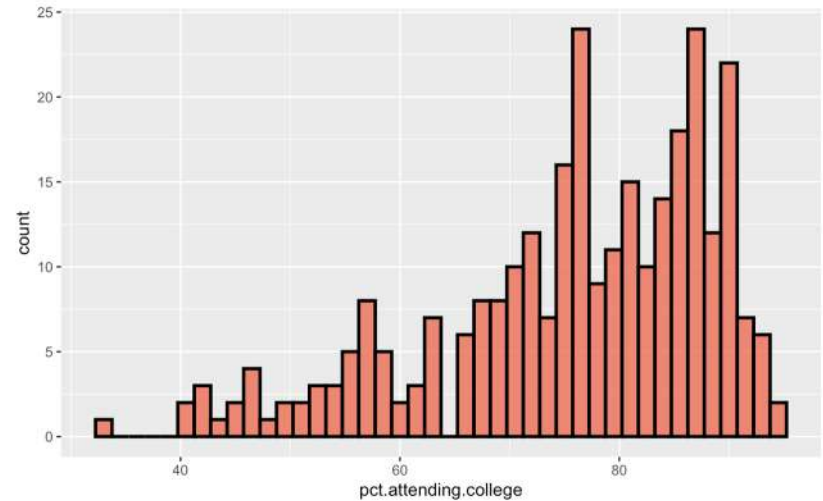


5. Average Teacher Salary

- Outliers
- Bell curve

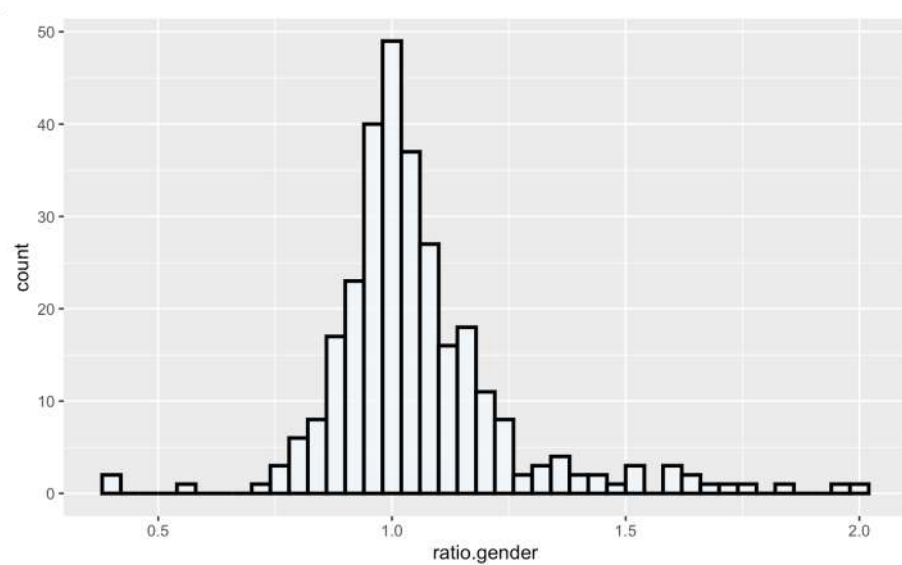
6. % of People Attending College

- Skewed to the left

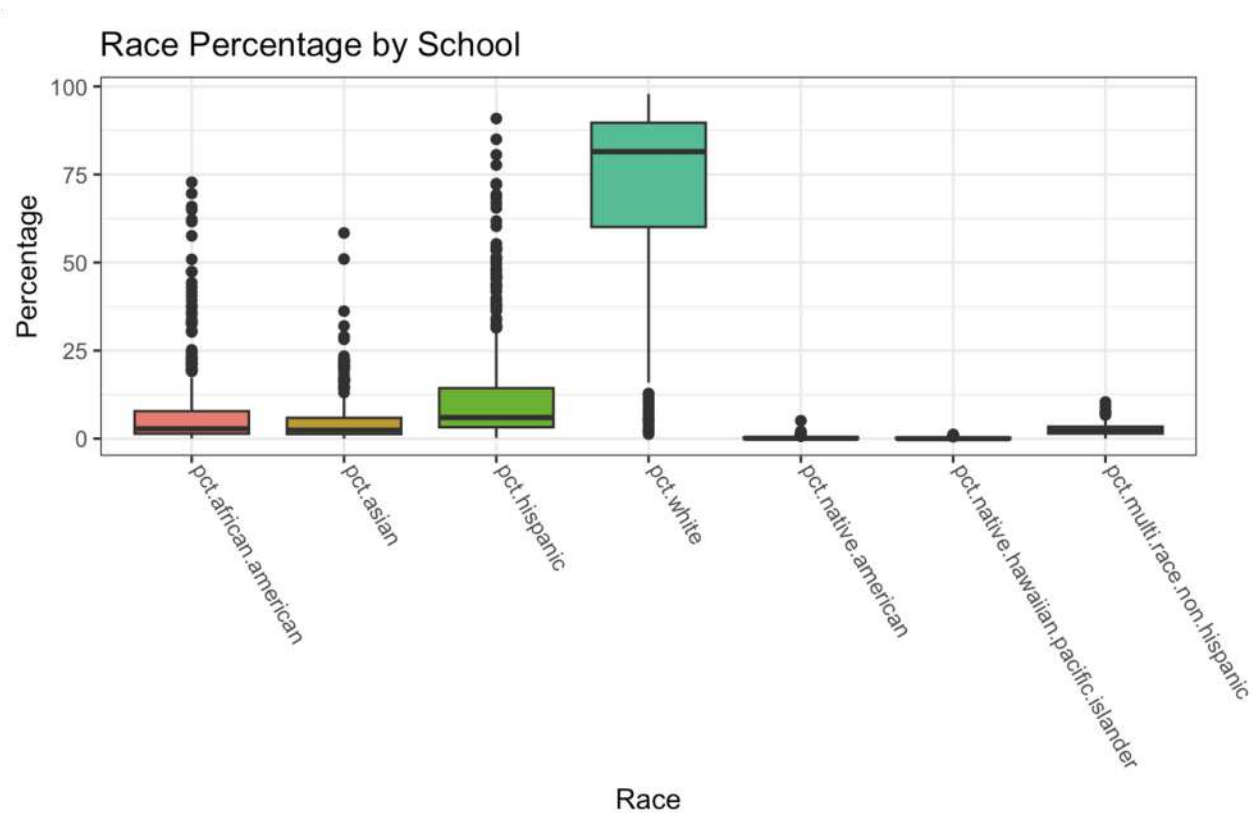


7. Gender Ratio

- Ratio of boys to girls
- Normal Curve



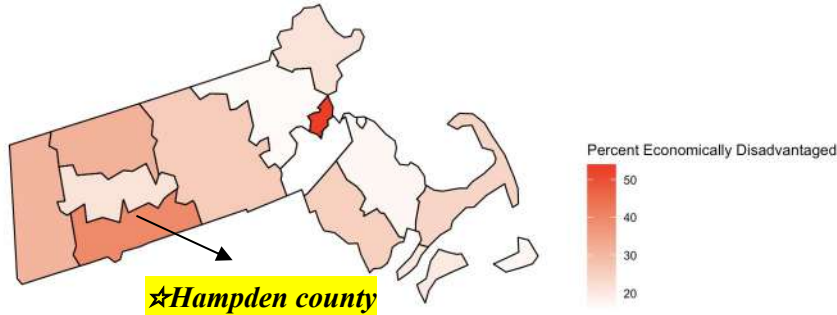
8. Ethnicity Ratio



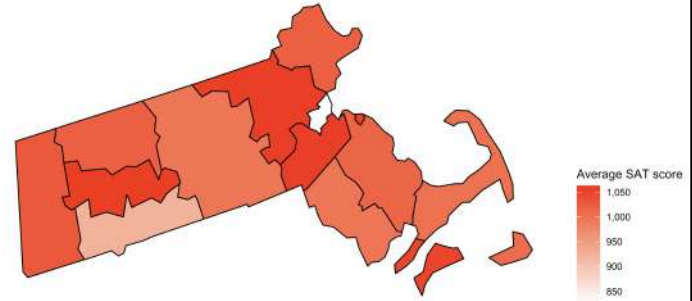
9. Heat Map

= Shows correlation and outlier

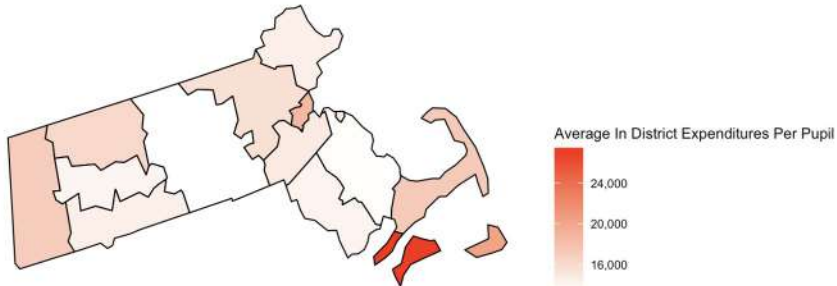
Percent Economically Disadvantaged by County
Massachusetts



Average SAT score by County
Massachusetts



Average In District Expenditures Per Pupil
Massachusetts



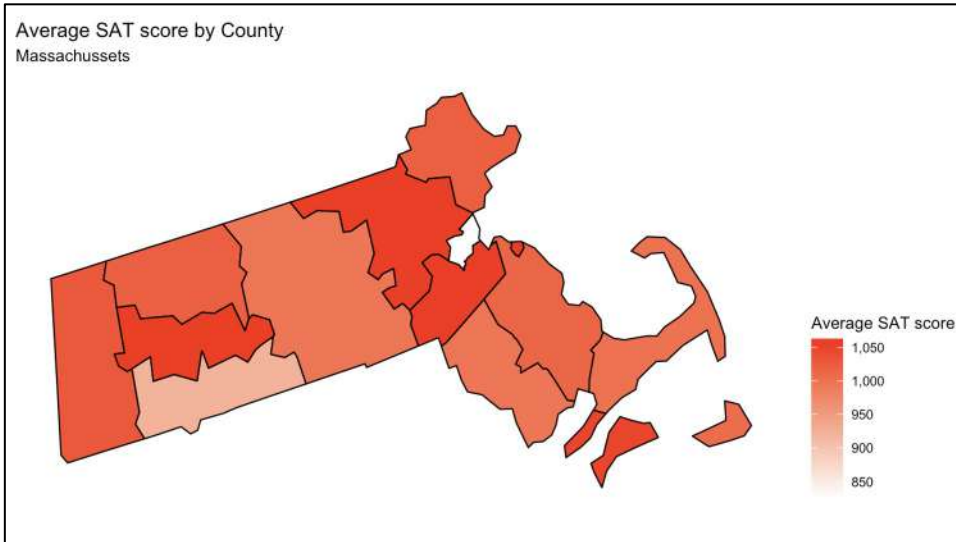
Interpretation example:

Hampden county:

The area which has high percent economically disadvantaged and has low average district expenditure has low average SAT score in the area.

9. Heat Map

= Boston as an outlier



Why is Boston an outlier for public schools?

= Low quality education

“ The district has failed to effectively serve its most vulnerable students, carry out basic operational functions, and address systemic barriers to providing an equitable, quality education. ”

3. OUR MODEL

Models

1

**Multiple Linear
Regression**

2

**Relaxed
LASSO**

3

Decision Tree

1. Multiple Linear Regression

Residuals:

Min	1Q	Median	3Q	Max
-147.47	-25.96	-0.99	28.22	205.56

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	2294.9168657	3576.2119954	0.64	0.52
pct.first.language.not.english	-0.1089156	0.4058403	-0.27	0.79
pct.students.with.disabilities	-0.0512011	0.6046603	-0.08	0.93
pct.economically.disadvantaged	-3.4055061	0.3882995	-8.77	< 0.0000000000000002 ***
pct.african.american	-15.8858968	35.7866550	-0.44	0.66
pct.asian	-11.8430955	35.8310369	-0.33	0.74
pct.hispanic	-14.5767707	35.7874113	-0.41	0.68
pct.white	-14.2458825	35.7589126	-0.40	0.69
pct.native.american	-30.7847171	35.4550348	-0.87	0.39
pct.native.hawaiian.pacific.islander	-36.9849226	39.3112521	-0.94	0.35
pct.multi.race.non.hispanic	-1.2091804	35.8155705	-0.03	0.97
ratio.gender	-10.7775011	19.3258689	-0.56	0.58
average.teacher.salary	0.0002921	0.0004442	0.66	0.51
average.in.district.expenditures.per.pupil	0.0000929	0.0011725	0.08	0.94
pct.attending.college	2.4532730	0.3867793	6.34	0.0000000009 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 46.2 on 280 degrees of freedom

Multiple R-squared: 0.854, Adjusted R-squared: 0.847

F-statistic: 117 on 14 and 280 DF, p-value: <0.0000000000000002

1.1 Multiple Linear Regression with Backwards elimination

```
Call:
lm(formula = sum.SAT ~ pct.economically.disadvantaged + pct.african.american +
    pct.asian + pct.hispanic + pct.white + pct.native.american +
    pct.attending.college, data = data.final)

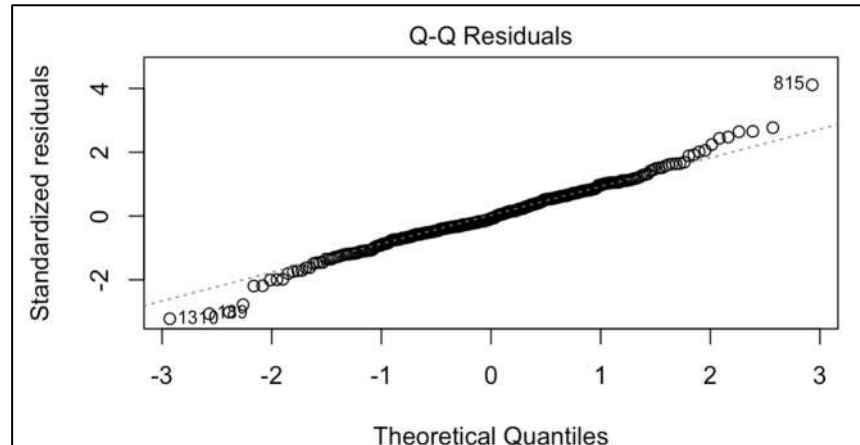
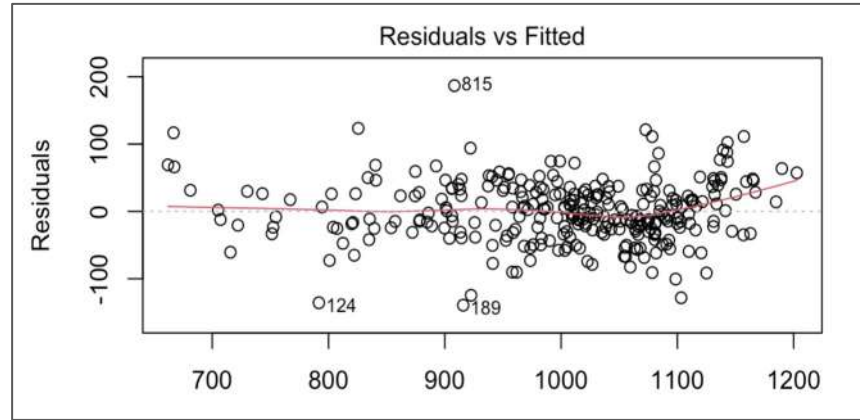
Residuals:
    Min       1Q   Median       3Q      Max
-210.12  -42.48   -4.16   42.64  282.81

Coefficients:
                Estimate Std. Error t value      Pr(>|t|)
(Intercept)      3276.957     279.692    11.72 < 0.0000000000000002 ***
pct.economically.disadvantaged -5.219       0.534    -9.77 < 0.0000000000000002 ***
pct.african.american  -22.287       2.857    -7.80 0.00000000000001147 ***
pct.asian          -16.532       3.002    -5.51 0.00000000807484879 ***
pct.hispanic       -20.310       2.784    -7.30 0.0000000000029214 ***
pct.white          -19.749       2.846    -6.94 0.0000000000262969 ***
pct.native.american -44.712      11.385    -3.93      0.00011 ***
pct.attending.college   3.803       0.453     8.39 0.0000000000000022 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 69.8 on 287 degrees of freedom
Multiple R-squared:  0.85,    Adjusted R-squared:  0.847
F-statistic: 233 on 7 and 287 DF, p-value: <0.0000000000000002
```

$$\hat{Y} = 3331.242 - (5.186 * PctEconomicallyDisadvantaged) - (22.805 * PctAfricanAmerican) - (17.219 * PctAsian) - (20.933 * PctHispanic) \\ - (44.576 * PctNativeAmerican) - (59.209 * PctNativeHawaiianPacificIslander) + (3.927 * PctAttendingCollege)$$

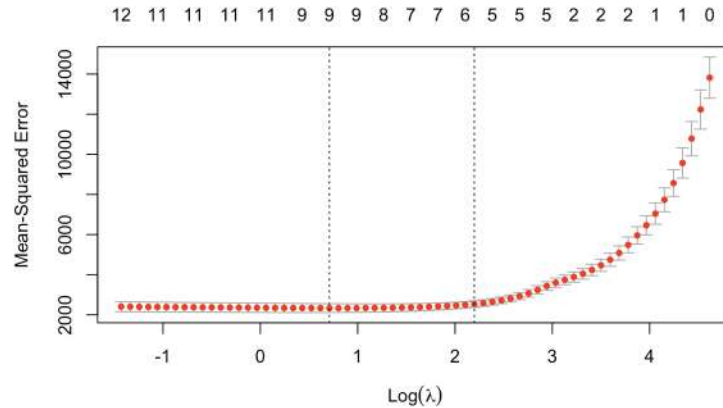
1.2 Multiple Linear Regression: Linearity and Homoscedasticity



Confidence Interval

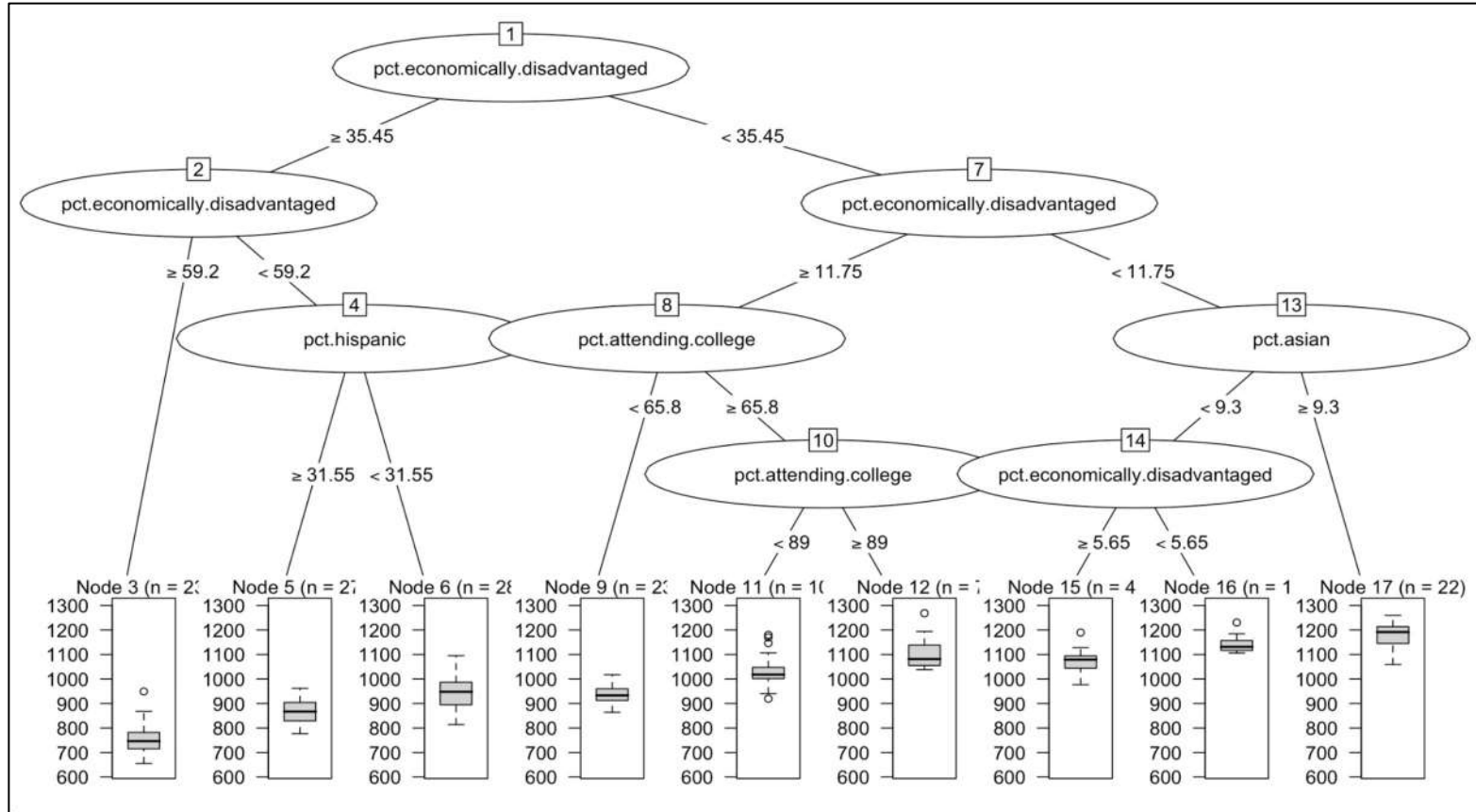
	2.5 %	97.5 %
(Intercept)	820.39	926.42
pct.economically.disadvantaged	-4.33	-3.32
pct.african.american	-2.11	-1.03
pct.asian	1.66	3.22
pct.multi.race.non.hispanic	9.22	16.31
pct.attending.college	1.95	3.12

2. Relaxed LASSO



	(Intercept)	pct.economically.disadvantaged	pct.african.american
pct.asian	1336.02	-5.30	-2.27
2.90			
	pct.hispanic	pct.native.american	pct.native.hawaiian.pacific.islander
pct.multi.race.non.hispanic	-0.49	-17.31	-20.86
18.24			
	pct.attending.college	ratio.gender	
	3.61	-12.55	
[1]	"(Intercept)"	"pct.economically.disadvantaged"	"pct.african.american"
[4]	"pct.asian"	"pct.hispanic"	"pct.native.american"
[7]	"pct.native.hawaiian.pacific.islander"	"pct.multi.race.non.hispanic"	"pct.attending.college"
[10]	"ratio.gender"		

3. Decision Tree



4. FINDINGS & CONCLUSION

Conclusions

Findings:

- 1. What are variables affecting SAT score?*
- 2. Is SAT biased?*

Generalization:

What is the largest population we can generalize our data to?

Limitations of our findings

Implications:

What can be learn from our data?

Findings

Variables correlated to SAT scores

(relaxed lasso)

```
Call:
lm(formula = sum.SAT ~ pct.economically.disadvantaged + pct.african.american +
    pct.asian + pct.multi.race.non.hispanic + pct.attending.college,
    data = data.final2)
```

Residuals:

Min	1Q	Median	3Q	Max
-196.94	-42.01	-3.83	43.03	290.27

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	1296.208	41.017	31.60	< 0.0000000000000002 ***
pct.economically.disadvantaged	-5.725	0.391	-14.65	< 0.0000000000000002 ***
pct.african.american	-2.454	0.418	-5.88	0.00000001155537437 ***
pct.asian	3.229	0.605	5.33	0.00000019373479498 ***
pct.multi.race.non.hispanic	19.814	2.744	7.22	0.00000000000459748 ***
pct.attending.college	3.884	0.451	8.62	0.0000000000000044 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 70.2 on 289 degrees of freedom

Multiple R-squared: 0.847, Adjusted R-squared: 0.845

F-statistic: 321 on 5 and 289 DF, p-value: <0.0000000000000002

Negative correlation:

- Percent economically disadvantaged
- Percent african american

Variables correlated to SAT scores

(relaxed lasso)

```
Call:
lm(formula = sum.SAT ~ pct.economically.disadvantaged + pct.african.american +
    pct.asian + pct.multi.race.non.hispanic + pct.attending.college,
    data = data.final2)
```

Residuals:

Min	1Q	Median	3Q	Max
-196.94	-42.01	-3.83	43.03	290.27

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	1296.208	41.017	31.60	< 0.0000000000000002 ***
pct.economically.disadvantaged	-5.725	0.391	-14.65	< 0.0000000000000002 ***
pct.african.american	-2.454	0.418	-5.88	0.00000001155537437 ***
pct.asian	3.229	0.605	5.33	0.00000019373479498 ***
pct.multi.race.non.hispanic	19.814	2.744	7.22	0.00000000000459748 ***
pct.attending.college	3.884	0.451	8.62	0.0000000000000044 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 70.2 on 289 degrees of freedom
Multiple R-squared: 0.847, Adjusted R-squared: 0.845
F-statistic: 321 on 5 and 289 DF, p-value: <0.0000000000000002

Positive correlation:

- Percentage of asians
- Percentage of multi-race
- Percentage of alumnus attending college

Findings:

Ethnicity, Socio-economic status, and education quality affects SAT scores

Is the SAT biased?

Yes. Here's why.

What causes score disparities?

```
Call:
lm(formula = sum.SAT ~ pct.economically.disadvantaged + pct.african.american +
    pct.asian + pct.multi.race.non.hispanic + pct.attending.college,
    data = data.final2)
```

Residuals:

Min	1Q	Median	3Q	Max
-196.94	-42.01	-3.83	43.03	290.27

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	1296.208	41.017	31.60	< 0.0000000000000002 ***
pct.economically.disadvantaged	-5.725	0.391	-14.65	< 0.0000000000000002 ***
pct.african.american	-2.454	0.418	-5.88	0.0000000115537437 ***
pct.asian	3.229	0.605	5.33	0.00000019373479498 ***
pct.multi.race.non.hispanic	19.814	2.744	7.22	0.00000000000459748 ***
pct.attending.college	3.884	0.451	8.62	0.0000000000000044 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 70.2 on 289 degrees of freedom
Multiple R-squared: 0.847, Adjusted R-squared: 0.845
F-statistic: 321 on 5 and 289 DF, p-value: <0.0000000000000002

Racial gaps

Socio- economic gaps

Education quality gaps

Generalization

Application of our data

(Which populations can our data be applied to?)

1. Not applicable to for all US schools public schools

2. But, applicable to states with similar demographics with
Massachusetts (Ontology heat map)

Generalization Scope

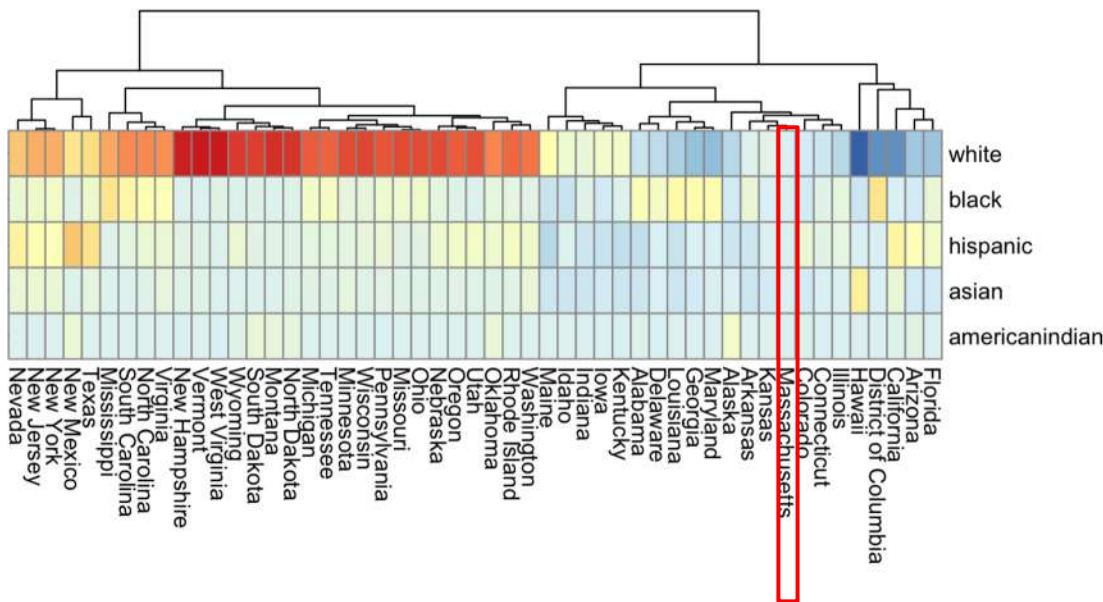
	US average	Our dataset
Percent economically disadvantaged	35	25.6
Percent African American	16	8.15
Percent Asian	5	5
Percent Hispanic	23	14.3
Percent White	52	69.6
Percent Native American	1	0.224
Percent Attending College	70	75.8

Average SAT score	1050	1000
-------------------	------	------

1. Cannot be generalized to all US schools.

= The dataset is likely to underestimate SAT scores if generalized to entire US due to varying demographic.

Generalization Scope: Ontology Heatmap



2. Can be generalized to public schools in states with similar demographics as Massachusetts

- Illinois
- Connecticut
- Colorado
- Kansas
- Arkansas
- Alaska
- Kentucky
- Iowa
- Indiana
- Idaho

*Created from dataset "Racepct_bystate_2017.csv"
from governing.com*

Limitations

Limitations

**Cannot be
generalized to
inapplicable US
states.**

**Cannot be
generalized to
private schools.**

Limitations shown in real life scenario

School type	Location	Actual Average SAT score	Predicted Average SAT score using our model
All-girls private	Texas	1450	1150
Co-ed private	Virginia	1330	1150
Co-ed Public	Massachusetts	1240	1200
Co-ed Public	Illinois	1180	1200

Implications:

What can be learned from our data?

Implications

**1. SAT is not a
standardized and
impartial exam**

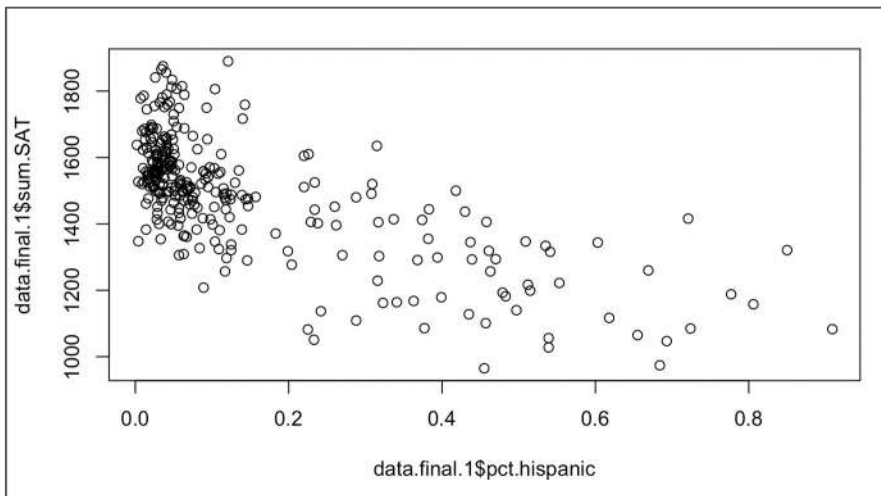
**2. SAT preparation
resources should
be more
accessible**

THANK YOU

Questions?

APPENDIX

Appendix 1. High intercept explanation



```
Call:
lm(formula = sum.SAT ~ pct.african.american + pct.asian + pct.white +
    pct.native.american + pct.economically.disadvantaged + pct.attending.college,
    data = data.final.1)
```

```
Residuals:
    Min       1Q   Median       3Q      Max
-268.69  -47.85   -7.01   41.49  302.42
```

```
Coefficients:
              Estimate Std. Error t value      Pr(>|t|)
(Intercept)    1284.1       65.3    19.66 < 0.0000000000000002 ***
pct.african.american -179.5       56.7     -3.17      0.0017 **
pct.asian       472.4       78.4      6.02    0.0000000052068 ***
pct.white       82.6       41.6      1.99      0.0480 *
pct.native.american -800.7    1110.1     -0.72      0.4713
pct.economically.disadvantaged -507.5      58.0    -8.74 < 0.0000000000000002 ***
pct.attending.college 359.1      49.2      7.31    0.0000000000027 ***
---

```

```
Call:
lm(formula = sum.SAT ~ pct.african.american + pct.asian + pct.white +
    pct.native.american + pct.economically.disadvantaged + pct.attending.college +
    pct.hispanic, data = data.final.1)
```

```
Residuals:
    Min       1Q   Median       3Q      Max
-210.12  -42.48   -4.16   42.64  282.81
```

```
Coefficients:
              Estimate Std. Error t value      Pr(>|t|)
(Intercept)    3277.0       279.7    11.72 < 0.0000000000000002 ***
pct.african.american -2228.7       285.7    -7.80    0.000000000001147 ***
pct.asian       -1653.2       300.2    -5.51    0.00000000807484879 ***
pct.white       -1974.9       284.6    -6.94    0.000000000262969 ***
pct.native.american -4471.2    1138.5    -3.93      0.00011 ***
pct.economically.disadvantaged -521.9      53.4    -9.77 < 0.0000000000000002 ***
pct.attending.college 380.3      45.3      8.39    0.000000000000022 ***
pct.hispanic    -2031.0       278.4    -7.30    0.00000000029214 ***

```

Variables correlated to SAT scores by Multiple Linear Regression

```
Call:
lm(formula = sum.SAT ~ pct.economically.disadvantaged + pct.african.american +
  pct.asian + pct.hispanic + pct.white + pct.native.american +
  pct.attending.college, data = data.final)
```

```
Residuals:
    Min       1Q   Median       3Q      Max
-210.12  -42.48   -4.16   42.64  282.81
```

```
Coefficients:
```

	Estimate	Std. Error	t value
Pr(> t)			
(Intercept)	3276.957	279.692	11.72 <
0.0000000000000002 ***			
pct.economically.disadvantaged	-5.219	0.534	-9.77 <
0.0000000000000002 ***			
pct.african.american	-22.287	2.857	-7.80
0.0000000000001147 ***			
pct.asian	-16.532	3.002	-5.51
0.0000000000007484879 ***			
pct.hispanic	-20.310	2.784	-7.30
0.00000000000029214 ***			
pct.white	-19.749	2.846	-6.94
0.000000000000262969 ***			
pct.native.american	-44.712	11.385	-3.93
0.00011 ***			
pct.attending.college	3.803	0.453	8.39
0.0000000000000022 ***			

```
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
Residual standard error: 69.8 on 287 degrees of freedom
Multiple R-squared:  0.85,    Adjusted R-squared:  0.847
F-statistic: 233 on 7 and 287 DF,  p-value: <0.0000000000000002
```

= If the percentage of students who are economically disadvantaged of a given school is increased, SAT scores decrease by 5.219. (If all other variables are controlled)

→ Shows correlation between socioeconomic status and SAT scores (i.e. the economically advantaged is more likely to have higher scores)

Variables correlated to SAT scores by Multiple Linear Regression

```
Call:
lm(formula = sum.SAT ~ pct.economically.disadvantaged + pct.african.american +
    pct.asian + pct.hispanic + pct.white + pct.native.american +
    pct.attending.college, data = data.final)
```

Residuals:

Min	1Q	Median	3Q	Max
-210.12	-42.48	-4.16	42.64	282.81

Coefficients:

	Estimate	Std. Error	t value
Pr(> t)			
(Intercept)	3276.957	279.692	11.72 <
0.0000000000000002 ***			
pct.economically.disadvantaged	-5.219	0.534	-9.77 <
0.0000000000000002 ***			
pct.african.american	-22.287	2.857	-7.80
0.0000000000001147 ***			
pct.asian	-16.532	3.002	-5.51
0.0000000000007484879 ***			
pct.hispanic	-20.310	2.784	-7.30
0.00000000000029214 ***			
pct.white	-19.749	2.846	-6.94
0.000000000000262969 ***			
pct.native.american	-44.712	11.385	-3.93
0.00011 ***			
pct.attending.college	3.803	0.453	8.39
0.0000000000000022 ***			

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1			

Residual standard error: 69.8 on 287 degrees of freedom
Multiple R-squared: 0.85, Adjusted R-squared: 0.847
F-statistic: 233 on 7 and 287 DF, p-value: <0.0000000000000002

= For each ethnicity, if their percentage is increased, SAT scores all decrease. (If all other variables are controlled)

*** Comparing scores, noting that Asians has the smallest estimated values it shows that Asians most likely to have higher SAT scores, while Native Americans are least likely to have high SAT scores.*

Variables correlated to SAT scores by Multiple Linear Regression

```
Call:
lm(formula = sum.SAT ~ pct.economically.disadvantaged + pct.african.american +
  +   pct.asian + pct.hispanic + pct.white + pct.native.american +
  +   pct.attending.college, data = data.final)
```

```
Residuals:
    Min       1Q   Median       3Q      Max
-210.12  -42.48   -4.16   42.64  282.81
```

```
Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)    3276.957    279.692   11.72 < 0.0000000000000002 ***
pct.economically.disadvantaged -5.219      0.534  -9.77 < 0.0000000000000002 ***
pct.african.american    -22.287      2.857   -7.80 0.0000000000001147 ***
pct.asian    -16.532      3.002   -5.51 0.00000000807484879 ***
pct.hispanic    -20.310      2.784   -7.30 0.00000000000029214 ***
pct.white    -19.749      2.846   -6.94 0.000000000000262969 ***
pct.native.american    -44.712     11.385   -3.93 0.00011 ***
pct.attending.college      3.803      0.453    8.39 0.0000000000000022 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
Residual standard error: 69.8 on 287 degrees of freedom
Multiple R-squared:  0.85,    Adjusted R-squared:  0.847
F-statistic: 233 on 7 and 287 DF, p-value: <0.0000000000000002
```

= If the percentage of alumnus of a given school attending college is increased, SAT scores increase by 3.803. (If all other variables are controlled)

→ High percentage of alumnus attending college denotes strong academic background of a given school.